

Math 271: Mathematical Methods (Semester I, 2026–27)

Class Test 4 (21 September 2026) — Comprehensive Marking Scheme

General principles

- G1.** Marks are awarded for *logically justified* steps. A correct statement that is asserted without the reason that makes it true earns at most about half of the marks for that step.
- G2.** Every mathematically valid route gets full credit. The main alternatives seen in the scripts are listed under each part, and each is mapped onto the same step allocation.
- G3.** Ordinary slips (a strict inequality where $\leq$ is needed, $\lfloor H \rfloor + 1$ written as $H + 1$ for real H , writing x_n for $f(x_n)$ once, and so on) are not penalised.
- G4.** A *false* mathematical claim that the argument depends on (e.g. “if a subsequence converges then the whole sequence converges”, “closed $\Rightarrow$ every sequence converges”, “ $\sup S \in S$ ”) loses the marks for every step that rests on it.
- G5.** Assuming the conclusion (e.g. “let $x_M \in C$ be such that $f(x_M) = \sup f(C)$ ” before proving it exists) earns no marks for the steps it replaces.
- G6.** In (c) and (d), students may quote the Bolzano–Weierstrass theorem in the form “every sequence in a compact (closed and bounded) $C \subseteq \mathbb{R}^n$ has a subsequence converging to a point of C ”, as well as results (a), (b), (c) of this test.

1 Part (a) [15 points]

Claim. If $\emptyset \neq S \subseteq \mathbb{R}$ is bounded above, some sequence in S converges to $\sup S$.

Model solution (Route A1: $M - \frac{1}{n}$ is not an upper bound)

- a1.** By the Continuum Property, $M = \sup S$ exists, and $x \leq M$ for all $x \in S$. [2 points]
- a2.** Fix $n \in \mathbb{N}$. Since $M - \frac{1}{n} < M$ and M is the *least* upper bound, $M - \frac{1}{n}$ is *not an upper bound* of S . Hence there is $x_n \in S$ with $x_n > M - \frac{1}{n}$. Doing this for every n gives a sequence $\{x_n\} \subseteq S$. [5 points]
 - 5: existence of x_n explicitly derived from “ $M - \frac{1}{n}$ is not an upper bound”.
 - 3: existence of x_n asserted with no reason, or with a wrong reason (“by the Archimedean property”, “ $M - \frac{1}{n}$ is not the *least* upper bound”).
 - 1: a false version of the supremum property (e.g. “ $M - \varepsilon \in S$ ”).
- a3.** Since M is an upper bound, $x_n \leq M$; thus $M - \frac{1}{n} < x_n \leq M$, i.e. $|x_n - M| < \frac{1}{n}$. [4 points]
 - -1: writing $x_n < M$ (strict) or “ $x < M$ for all $x \in S$ ”; the supremum may belong to S .
 - -1: getting $|x_n - M| < \frac{1}{n}$ from the upper estimate $x_n < M + \frac{1}{n}$ alone.
- a4.** Given $\varepsilon > 0$, use the Archimedean property to choose $N \in \mathbb{N}$ with $N > 1/\varepsilon$. [2 points]
- a5.** Then for every $n \geq N$: $|x_n - M| < \frac{1}{n} \leq \frac{1}{N} < \varepsilon$, so $x_n \rightarrow M = \sup S$. [2 points]

Alternative valid routes

Route A2 (general ε , then $\varepsilon_n = 1/n$).

First prove “for every $\varepsilon > 0$ there is $x \in S$ with $M - \varepsilon < x \leq M$ ”, then put $\varepsilon = 1/n$. Marked exactly as A1.

Route A3 (Squeeze/Sandwich theorem).

Starting from $M - \frac{1}{n} < x_n \leq M$, apply the squeeze theorem with $M - \frac{1}{n} \rightarrow M$ and the constant sequence $M \rightarrow M$. a4+a5 (4 marks) are given for a correct squeeze argument. -0.5 if the student writes $\lim x_n$ inside the inequality before its existence is known, without citing the squeeze theorem as the reason it exists. -1 if a strict inequality is kept after taking limits ($\lim(M - \frac{1}{n}) < \lim x_n$).

Route A4 (case split).

If $M \in S$, the constant sequence $x_n = M$ works. If $M \notin S$, use A1. The case split is not needed but is valid. Credit follows A1 for the second case.

Invalid approaches (illustrative)

- “Any increasing convergent sequence in S converges to $\sup S$ ” is false: take $S = [0, 1]$ and $x_n = \frac{1}{2} - \frac{1}{n}$. This gets at most a1 (2 marks).
- Asserting “given ε_n there is x_n with $|x_n - M| < \varepsilon_n$ ” with no link to the supremum property and no choice of $\varepsilon_n \rightarrow 0$ gets at most a1 plus 2.

2 Part (b) [15 points]

Claim. If $\emptyset \neq S \subseteq \mathbb{R}$ is unbounded *above*, some sequence in S diverges to $+\infty$.

Model solution

b1. Correct use of the hypothesis. “ S is not bounded above” means: for every $K \in \mathbb{R}$ there is $x \in S$ with $x > K$. [3 points]

- Using instead the negation of *bounded* ($|x| > K$) gets 1 mark. It only produces $|x_n| > n$, which does *not* imply $x_n \rightarrow +\infty$, since the terms could tend to $-\infty$.

b2. Construction. With $K = n$, choose $x_n \in S$ with $x_n > n$ for each $n \in \mathbb{N}$. This defines a sequence in S . [6 points]

- 3 if the chosen terms satisfy only $|x_n| > n$; 0–1 if the “sequence” is not in S (e.g. $x_n = nx_0$).

b3. Verification against the definition. Given $H > 0$, choose $N \in \mathbb{N}$ with $N \geq H$ (e.g. $N = \lfloor H \rfloor + 1$, by the Archimedean property). Then for $n > N$: $x_n > n > N \geq H$. [4 points]

- 1–2 for “ $x_n > n \rightarrow \infty$ hence $x_n \rightarrow \infty$ ” with no H – N argument; -0.5 for $N = H + 1$ with H real (not an integer) or for a dropped quantifier.

b4. Conclusion stated in terms of the given definition of divergence to $+\infty$. [2 points]

Alternative valid routes

- **Monotone construction:** choose $x_{n+1} \in S$ with $x_{n+1} > \max\{x_n, n+1\}$. Valid, same credit.
- **Restricting to positive elements** (“WLOG $x > 0$ ”) is unnecessary but harmless.
- **Phrasing via limits** (“ $\lim x_n \geq \lim n = +\infty$ ”) shows the right idea but is not a proof from the stated definition: b3 gets at most 1.

3 Part (c) [35 points]

Claim. If $C \subseteq \mathbb{R}^n$ is compact and $f : C \rightarrow \mathbb{R}$ is continuous, then f is bounded on C .

Model solution (Route C1: contradiction via Bolzano–Weierstrass)

c1. Set-up and construction. Suppose f is not bounded. Then for every $n \in \mathbb{N}$ there is $x_n \in C$ with $|f(x_n)| > n$. (Equivalently: if f is not bounded above, choose $f(x_n) > n$. Here (b) may be applied to $f(C)$.) [6 points]

- 3–4 if the existence of a sequence with $f(x_n) \rightarrow \infty$ is merely asserted, or if the points x_n in C are never identified.

c2. Compactness (Bolzano–Weierstrass). $\{x_n\} \subseteq C$ and C is bounded, so $\{x_n\}$ has a convergent subsequence $x_{n_k} \rightarrow x^*$. Since C is closed, $x^* \in C$. [10 points]

- 7 for the convergent subsequence (BW, using boundedness); 3 for $x^* \in C$ (closedness), which is needed to apply continuity at x^* .
- A false claim such as “ $x_n \rightarrow x$ because C is closed”, or “if a subsequence converges then the whole sequence converges”, gets 0–3 for this step.

c3. Continuity (sequential criterion). f is continuous at $x^* \in C$, so $f(x_{n_k}) \rightarrow f(x^*) \in \mathbb{R}$. [6 points]

c4. The contradiction (rigorous). [13 points]

- (i) Explain why c3 is incompatible with c1: [8]
 $|f(x_{n_k})| > n_k \geq k$, so $\{f(x_{n_k})\}$ is unbounded, whereas every convergent sequence is bounded. Equivalently: a subsequence of a sequence diverging to $+\infty$ also diverges to $+\infty$ and cannot converge. (5–6 if the contradiction is just declared, e.g. “ $f(x_{n_k})$ is convergent, contradiction”; 4–5 if ∞ is treated as a number, e.g. “ $\lim f(x_{n_k}) > \infty$ ”.)
- (ii) Both directions covered: use of $|f|$, or a separate “unbounded below” case, or $g = -f$. [3]
 (2 for a one-line “similarly”; 0 if the below case is ignored and only $f(x_n) > n$ is used.)
- (iii) Clear conclusion: the assumption fails, so f is bounded on C . [2]

Alternative valid routes**Route C2 (via part (b)).**

If $f(C)$ is unbounded above, part (b) gives $y_n = f(x_n) \in f(C)$ with $y_n \rightarrow +\infty$. Take preimages $x_n \in C$, then apply c2–c4. Using (b) earns c1 in full.

Route C3 (explicit bound on the subsequence).

After $f(x_{n_k}) \rightarrow f(x^*)$, take $\varepsilon = 1$: there is K with $|f(x_{n_k})| < 1 + |f(x^*)|$ for $k \geq K$. Then $B = \max\{|f(x_{n_1})|, \dots, |f(x_{n_{K-1}})|, 1 + |f(x^*)|\}$ bounds the whole subsequence, contradicting $|f(x_{n_k})| > n_k \rightarrow \infty$. This is a fully rigorous version of c4(i).

Route C4 (convergent $\Rightarrow$ bounded).

$f(x_{n_j})$ converges, so $|f(x_{n_j})| \leq B$ for all j . Choosing $n_j > B$ gives $|f(x_{n_j})| > n_j > B$, a contradiction. Full credit.

Invalid approach

- Taking an *arbitrary* sequence in C , extracting a convergent subsequence, and concluding “ f converges, hence is bounded”. This never links the sequence to the unboundedness of f , so there is no proof. At most 5 for a correct statement of BW plus 2 for continuity.

4 Part (d) [35 points]

Claim. With C , f as in (c), there exist $x_m, x_M \in C$ such that $f(x_m) \leq f(x) \leq f(x_M)$ for all $x \in C$.

Model solution (Route D1: (c) + (a) + Bolzano–Weierstrass)

d1. Existence of the extreme values. By (c), $f(C) = \{f(x) : x \in C\}$ is non-empty and bounded. By the Continuum Property, $M = \sup f(C)$ and $m = \inf f(C)$ exist, and $m \leq f(x) \leq M$ for all $x \in C$. [5 points]

- The supremum must be of the *set of values* $f(C)$. “ $\sup C$ ” (the domain) is meaningless for $C \subseteq \mathbb{R}^n$ and is a conceptual error: 0–2 here, and the later steps are compromised.
- The supremum of $\{f(x_1), f(x_2), \dots\}$ for an *arbitrary* sequence is the wrong set: at most 2.

d2. Maximising sequence. By (a) applied to $f(C)$, there are $y_n \in f(C)$ with $y_n \rightarrow M$. Pick $x_n \in C$ with $f(x_n) = y_n$, so $f(x_n) \rightarrow M$. [7 points]

- –2 if the supremum is written as a function value (e.g. “ $f(x_1)$ is the supremum”) before it has been shown to be attained; –2 to –3 for confusing points of C with values (“ $z_n \rightarrow M$ in C ”).

d3. Compactness. By BW, $\{x_n\}$ has a subsequence $x_{n_k} \rightarrow x_M$, with $x_M \in C$ because C is closed. [7 points]

- –1 if $x_M \in C$ is not stated or justified; –2 if the subsequence step is skipped (“ $x_n \rightarrow x_M$ ”).

d4. Continuity. $f(x_{n_k}) \rightarrow f(x_M)$. [5 points]

d5. Identification of the limit. $\{f(x_{n_k})\}$ is a subsequence of $\{f(x_n)\}$, which converges to M , so it also converges to M . Limits are unique, so $f(x_M) = M \geq f(x)$ for all $x \in C$. [6 points]

d6. Minimum. Same argument with $m = \inf f(C)$ (the infimum analogue of (a)), or apply the maximum result to the continuous function $-f$: $-f(x) \leq -f(x_m) \Rightarrow f(x_m) \leq f(x)$. [5 points]

Alternative valid routes**Route D2 (inf and sup in parallel).**

Build two sequences, $f(x_n) \rightarrow m$ and $f(y_n) \rightarrow M$, and extract BW subsequences of both simultaneously. Full credit (d6 absorbed in the parallel argument).

Route D3 ($f(C)$ is closed and bounded).

$\sup f(C) \in f(C)$. If one *proves* that $f(C)$ is closed, i.e. that the limit of any convergent sequence $f(x_n)$ lies in $f(C)$ (BW + continuity, exactly d3–d5), then $M, m \in f(C)$. Full credit only if the closedness is proved. Saying “ $f(C)$ is closed by continuity” is *not* enough: continuous images of closed sets need not be closed, e.g. $f(x) = 1/x$ on $[1, \infty)$ has image $(0, 1]$. That case gets d1, d6 and at most 5 of d2–d5.